جامعة العلوم الاسلامية العالمية

The World Islamic Sciences and Education University

كلية تكنولوجيا المعلومات

Faculty of Information Technology

Project Title: Electricity Bill System

Course: Python

Instructor: Dr. Salah Alghyaline

1. Team Members:

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Code

JupyterLab Python 3 (ipykernel)

[1]:
```python
students = {
    "S001": {"name": "Ahmad Ali", "courses": {"CS101", "MATH201"}},
    "S002": {"name": "Sara Mahmoud", "courses": {"ENG150"}},
    "S003": {"name": "Omar Hussein", "courses": {"CS101", "ENG150", "PHYS110"}},
    "S004": {"name": "Lina Khaled", "courses": set()},
    "S005": {"name": "Yousef Nasser", "courses": {"CS101"}}
}

courses = {
    "CS101": {"name": "Introduction to Programming", "capacity": 3, "students": ["S001", "S003", "S005"]},
    "MATH201": {"name": "Calculus II", "capacity": 2, "students": ["S001"]},
    "ENG150": {"name": "Academic Writing", "capacity": 4, "students": ["S002", "S003"]},
    "PHYS110": {"name": "General Physics", "capacity": 2, "students": ["S003"]}
}
```

[2]:
```python
def add_student():
    student_id = input("Enter student ID: ")
    if student_id in students:
        print("Student ID already exists.")
    else:
        name = input("Enter student name: ")
        students[student_id] = {"name": name, "courses": set()}
        print("Student added successfully.")

def add_course():
    course_code = input("Enter course code: ")
    if course_code in courses:
        print("Course code already exists.")
    else:
        course_name = input("Enter course name: ")
        capacity = int(input("Enter course capacity: "))
        courses[course_code] = {"name": course_name, "capacity": capacity, "students": []}
        print("Course added successfully.")

def register_course():
    student_id = input("Enter student ID: ")
    course_code = input("Enter course code: ")
    if student_id not in students:
        print("Error: Student ID does not exist.")
```

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Code JupyterLab Python 3 (ipykernel)

```python
def register_course():
    student_id = input("Enter student ID: ")
    course_code = input("Enter course code: ")
    if student_id not in students:
        print("Error: Student ID does not exist.")
    elif course_code not in courses:
        print("Error: Course code does not exist.")
    elif course_code in students[student_id]["courses"]:
        print("Error: Student is already registered in this course.")
    elif len(courses[course_code]["students"]) >= courses[course_code]["capacity"]:
        print("Error: Course is full. Capacity reached.")
    else:
        students[student_id]["courses"].add(course_code)
        courses[course_code]["students"].append(student_id)
        print("Course registered successfully.")

def drop_course():
    student_id = input("Enter student ID: ")
    course_code = input("Enter course code: ")
    if student_id not in students:
        print("Error: Student ID does not exist.")
    elif course_code not in courses:
        print("Error: Course code does not exist.")
    elif course_code not in students[student_id]["courses"]:
        print("Error: Student is not enrolled in this course.")
    else:
        students[student_id]["courses"].remove(course_code)
        courses[course_code]["students"].remove(student_id)
        print("Course dropped successfully.")

def show_student_courses():
    student_id = input("Enter student ID: ")
    if student_id in students:
        print("Courses for", students[student_id]["name"], ":", students[student_id]["courses"])
    else:
        print("Error: Student ID does not exist.")

def show_course_students():
    course_code = input("Enter course code: ")
    if course_code in courses:
        print("Students in", courses[course_code]["name"], ":", courses[course_code]["students"])
```

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Code JupyterLab Python 3 (ipykernel)

```python
def show_course_students():
    course_code = input("Enter course code: ")
    if course_code in courses:
        print("Students in", courses[course_code]["name"], ":", courses[course_code]["students"])
    else:
        print("Error: Course code does not exist.")

def display_all_students():
    print("--- Students List ---")
    for s_id in students:
        print("ID:", s_id, "Name:", students[s_id]["name"], "Courses:", students[s_id]["courses"])

def display_all_courses():
    print("--- Courses List ---")
    for c_code in courses:
        print("Code:", c_code, "Name:", courses[c_code]["name"], "Capacity:", courses[c_code]["capacity"], "Registered:", len(courses[c_code]["students"
```

[3]:

```python
def main():
    while True:
        print("\n--- University Registration System ---")
        print("1. Add Student")
        print("2. Add Course")
        print("3. Register Course")
        print("4. Drop Course")
        print("5. Show Student Courses")
        print("6. Show Course Students")
        print("7. Display All Students")
        print("8. Display All Courses")
        print("9. Exit")

        choice = input("Enter your choice: ")

        if choice == "1":
            add_student()
        elif choice == "2":
            add_course()
        elif choice == "3":
            register_course()
        elif choice == "4":
            drop_course()
        elif choice == "5":
```

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Code JupyterLab Python 3 (ipykernel)

```python
        elif choice == "4":
            drop_course()
        elif choice == "5":
            show_student_courses()
        elif choice == "6":
            show_course_students()
        elif choice == "7":
            display_all_students()
        elif choice == "8":
            display_all_courses()
        elif choice == "9":
            print("Exiting...")
            break
        else:
            print("Invalid choice. Please try again.")

main()
```

```
--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course
4. Drop Course
5. Show Student Courses
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice:  2
Enter course code:  CS101
Course code already exists.

--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course
4. Drop Course
5. Show Student Courses
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice:  5
```

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Code JupyterLab Python 3 (ipykernel)

```
Enter your choice:  5
Enter student ID:  S004
Courses for Lina Khaled : set()

--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course
4. Drop Course
5. Show Student Courses
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice:  1
Enter student ID:  S006
Enter student name:  yousef ahmad
Student added successfully.

--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course
4. Drop Course
5. Show Student Courses
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice:  8
--- Courses List ---
Code: CS101 Name: Introduction to Programming Capacity: 3 Registered: 3
Code: MATH201 Name: Calculus II Capacity: 2 Registered: 1
Code: ENG150 Name: Academic Writing Capacity: 4 Registered: 2
Code: PHYS110 Name: General Physics Capacity: 2 Registered: 1

--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course
4. Drop Course
5. Show Student Courses
6. Show Course Students
7. Display All Students
```

```
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice:  9
Exiting...
```

[ ]: